

MERN Stack Technical Assessment

Position: Full Stack (MERN) Developer

DeadLine : 07/07/2026, 11:30 PM

Required Tech Stack

Backend

- Node.js
- Express.js
- TypeScript
- MongoDB
- Mongoose
- JWT Authentication

Frontend

- React
 - React Router
 - TypeScript
 - Tailwind CSS / Shadcn
 - Redux / tanstack/react-query
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Project

Mini ERP – Inventory & Sales Management System

Backend Requirements

Authentication

Implement JWT Authentication.

Features

- Login
 - Protected Routes
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Role Based Authorization

Implement the following roles:

- Admin
- Manager
- Employee

Permissions

Admin

- Full Access

Manager

- Manage Products
- Create Sales

Employee

- View Products
- Create Sales

Use proper authorization middleware for all protected routes.

Modules

Product Module

Fields

- Product Name
- SKU
- Category
- Purchase Price
- Selling Price
- Stock Quantity
- Product Image

Features

- CRUD Operations
- Image Upload during Product Creation/Update
- Search
- Pagination

Business Rules.

- Image upload is required while creating a product.
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Sales Module

Create Sales only.

Each Sale contains:

- Multiple Products
- Quantity

Backend Logic

- Automatically reduce stock.
 - Prevent selling unavailable or insufficient stock.
 - Calculate Grand Total.
 - Store Sale History.
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Dashboard API

Return the following statistics:

- Total Products
 - Total Sales
 - Low Stock Products (Stock < 5)
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Frontend Requirements

Authentication

- Login
 - Protected Routes
-

Dashboard

Display:

- Statistics Cards
 - Low Stock Products
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Products

- Product List
 - Add Product
 - Edit Product
 - Delete Product
 - Image Upload
 - Search
 - Pagination
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Sales

Create Sale Page with:

- Multiple Product Selection
 - Quantity Input
 - Automatic Total Calculation
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Validation & Best Practices

The project must include:

- Proper Error Handling
 - Input Validation
 - Protected APIs
 - Proper HTTP Status Codes
 - Consistent API Response Structure
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Bonus (Extra Marks)

Additional marks will be awarded for implementing any of the following:

- Dynamic Role & Permission Management (Database-driven roles and permissions)
 - Modular Feature-Based Architecture
 - Generic Query Builder (Search, Filter, Sort & Pagination)
 - Global Error Handler
 - Reusable functions
 - New technology implementation (e.g. socket.io / websocket)
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Submission Requirements

Please submit the following:

- Live Frontend URL
- Live Backend API URL
- Frontend GitHub Repository (Public)
- Backend GitHub Repository (Public)

- Admin Login Credentials
 - README (Project Setup & Installation Guide)
 - API Documentation (Backend)
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Late submissions may not be considered.